EFY Note
The source code of this project is available for free download at *source.efymag.com*

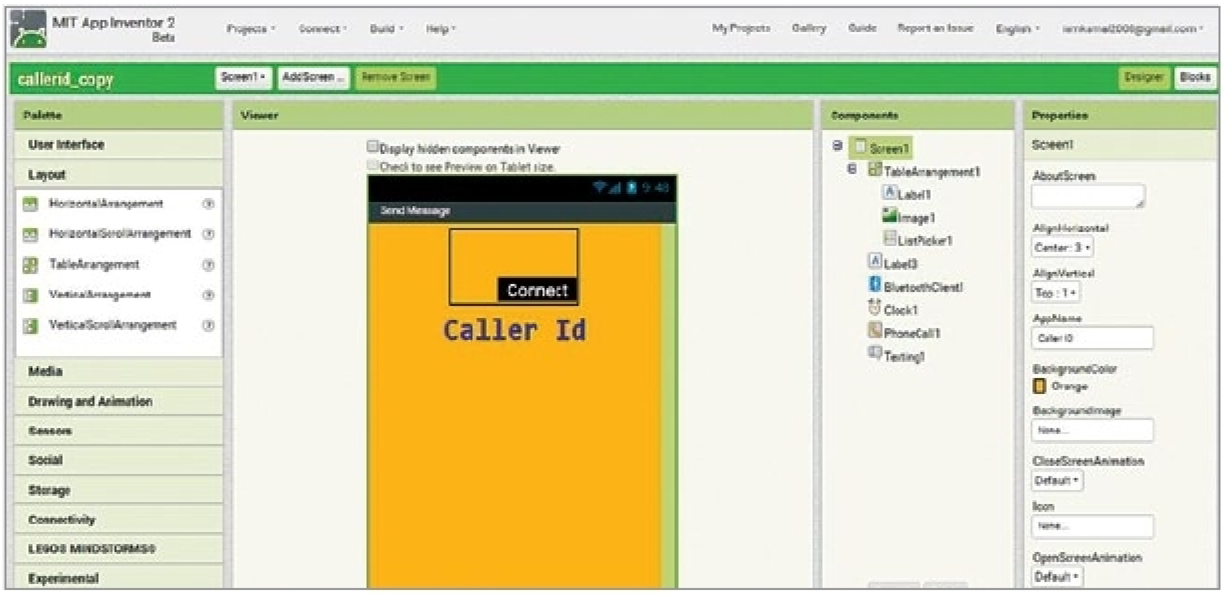

Fig. 3: Home screen of Android app in MIT App Inventor

connected to pins 5, 4, 3 and 2 of Arduino, respectively. Preset is used to adjust the contrast of the LCD.

Software for the wireless caller ID display is written in Arduino programming language. Arduino Uno is programmed using Arduino IDE. Select the correct board from Board > Tools menu in Arduino IDE. Select the correct COM port before uploading the program (caller_id.ino) through the standard USB port in the computer.

This project includes an Android application developed on MIT App Inventor platform. Block diagram of the app is shown in Fig. 2 and its home screen is shown in Fig. 3. Export this app as callerid.apk file, or with any other suitable file name. Install callerid.apk file on your smartphone. Note that this Android app is not available on Google Play Store.

Testing

Connect the LCD and Bluetooth module as shown in the circuit. Enclose the circuit properly in a suitable cabinet. Fix it near dashboard of your bike and power up the system.

To set up the caller ID system fitted near the dashboard of your bike, pair the Bluetooth device (HC-05) with your smartphone's Bluetooth. Then, open the callerid app and click on Connect to pair with the Bluetooth of the app. Make sure the Bluetooth module is connected to your smartphone's Bluetooth; else, it will not display any incoming call on the LCD.

You can now keep your smartphone in your pocket and drive safely. The LCD will show "Drive carefully" message when there is no incoming call. EFY

Kamal Das is an electronics hobbyist